

Robustness Evaluation of Autonomous Cyber Defense Agents under Adversarial Strategy Variation

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Abstract—Autonomous Cyber Defense (ACD) agents implementing Multi-Agent Reinforcement Learning (MARL) are increasingly studied for protecting enterprise networks. However, existing work evaluates agents using mean episodic reward under fixed attacker configurations, providing limited insight into policy robustness under adversarial variation. This paper evaluates the robustness of heuristic and graph-based MARL agents in the CybORG CAGE Challenge 4 environment. Agents are tested against multiple attackers and compared using a robustness score capturing average performance, within-strategy variability, and cross-strategy sensitivity. Reward and action decomposition analyze how policies obtain rewards and which actions they prioritize. Results show the heuristic agent achieves the highest robustness, maintaining stable performance across attacker strategies. The MARL agent exhibits higher variability and sensitivity to adversarial behavior. Behavioral analysis reveals that the policy relies heavily on connectivity-control actions while underutilizing analysis actions. These findings highlight challenges in learning robust defensive policies and demonstrate the importance of robustness-oriented evaluation for ACD.

Index Terms—Autonomous Cyber Defense, Robust AI, Multi-Agent Reinforcement Learning

I. INTRODUCTION

Autonomous Cyber Defense (ACD) systems implementing Multi Agent Reinforcement Learning (MARL) are increasingly explored for protecting complex enterprise and military networks. Recent work shows that graph-based MARL agents can learn decentralized defensive strategies in cyber environments such as Cyber Operations Research Gym (CybORG)/Cyber Autonomy Gym for Experimentation (CAGE) [1], [2]. These approaches typically evaluate agents using aggregate episodic reward as the primary performance metric.

However, high rewards under a fixed evaluation configuration does not necessarily imply robustness. In adversarial environments, policy performance can vary significantly across attacker strategies, stochastic events, and mission phases [3]. Evaluating agents solely by mean episodic reward therefore obscures important differences in stability and cross-strategy generalization.

In this work, we shift the focus from policy learning to robustness-oriented model comparison in the CAGE

Challenge 4 environment. We evaluate heuristic and Graph Neural Network (GNN)-based MARL cyber defense agents across multiple attacker strategies and introduce a robustness score that jointly captures expected performance, within-strategy variability, and cross-strategy sensitivity.

The main contributions of this paper are threefold:

- A robustness analysis of heuristic and GNN-based MARL cyber defense agents across heterogeneous attacker strategies.
- A robustness-oriented evaluation that explicitly accounts for stochastic instability and adversarial strategy variation.
- A composite robustness score enabling principled model selection beyond mean reward-based ranking.

The remainder of this paper is organized as follows. Section II reviews related work on ACD and robustness in Deep Reinforcement Learning (DRL). Section III describes the environment, evaluation setup, evaluated agents, and the robustness score. Section IV presents and analyzes the experimental results across attacker strategies. Finally, Section V concludes the paper and outlines directions for future work.

II. RELATED WORK

The cyber threat landscape is evolving rapidly due to advances in Artificial Intelligence (AI) and MARL, which enable increasingly automated and adaptive cyber operations [1], [2], [4], [5]. AI-driven agents are capable of automating cyber attack chains with near human-level effectiveness [6], thereby lowering the barrier for sophisticated adversarial operations [7]. At the same time, modern military and critical infrastructure networks are becoming increasingly complex, distributed, and mission-dependent [8]. These developments motivate the use of ACD systems that can autonomously detect, analyze, and mitigate cyber threats.

Recent research has explored MARL as a promising paradigm for ACD, particularly in simulation environments such as CybORG and the CAGE Challenge framework [9]. In these settings, network defense is typically modeled as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP), where a team of blue agents

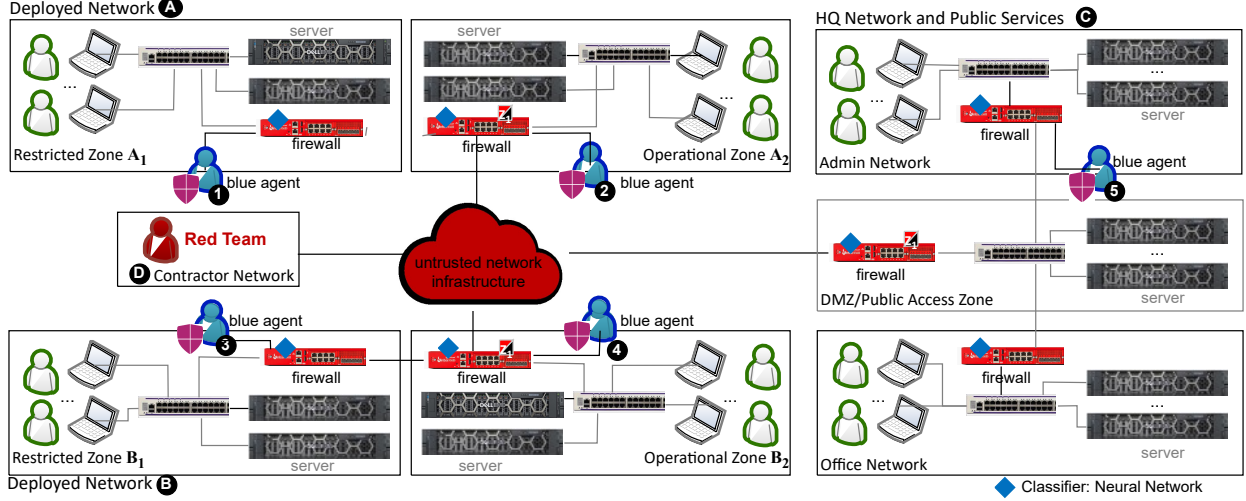


Fig. 1: Scenario with four networks (A-D). Deployed networks A and B each host two blue agents, while HQ network C hosts a fifth. Red agents hosted on contractor network D can use the interconnecting untrusted network infrastructure as entry point.

coordinate defensive actions under partial observability and communication constraints.

To represent complex enterprise networks, GNNs have been proposed as an effective learning architecture for processing graph-structured observations. Prior work [10] demonstrated that graph-based MARL agents can learn decentralized defensive policies in realistic enterprise network scenarios. However, most existing evaluations focus primarily on aggregate episodic reward under fixed attacker configurations [3], which provides limited insight into the robustness of learned policies under adversarial variation.

Explainability techniques have therefore been introduced to analyze the behavior of DRL agents in cyber defense environments. Methods such as SHapley Additive exPlanations (SHAP) [11] and reward decomposition [12] have been applied in the CybORG environment to attribute policy decisions to specific reward components and defensive actions [13]. These approaches provide valuable diagnostics for understanding learned strategies and identifying misalignment between reward signals and operational objectives.

More generally, robustness has emerged as a key challenge in DRL, particularly in adversarial or safety-critical environments where agents must maintain stable performance under stochastic disturbances and changing opponent behavior [14]. Existing approaches address robustness through techniques such as adversarial training and worst-case optimization [15]. In MARL settings, robustness is typically assessed by evaluating policy performance across diverse opponent strategies or environmental variations.

Despite these advances, robustness-oriented evaluation methodologies for decentralized MARL cyber defense agents remain limited. In particular, most studies rely on aggregate reward comparisons that do not explicitly capture variability across attacker strategies. To address this gap, we introduce a robustness score that jointly captures expected performance,

within-strategy variability, and cross-strategy sensitivity, enabling systematic comparison of heuristic and learning-based defense agents under heterogeneous adversarial behaviors.

III. METHODOLOGY

A. Environment and Evaluation Setup

All experiments are executed in an enterprise network scenario [3] (Fig. 1). The environment models a segmented enterprise network composed of multiple subnetworks $Z = \{z_1, \dots, z_k\}$ including Operational, Restricted, HQ/Admin, Contractor, and Public zones. The defended system contains a set of hosts, services, and network links forming a graph-structured infrastructure. Five distributed defensive agents (blue agents) jointly protect this network against adversarial agents (red agents) that attempt to discover hosts, escalate privileges, move laterally, and ultimately disrupt critical services. As illustrated by the scenario topology, the attacker initially operates from the contractor network and uses the interconnected infrastructure as an entry point into protected zones.

The environment is modeled as a Dec-POMDP. At each timestep t , the global environment state $s_t \in S$ captures the security state of all hosts, services, and connections. However, each blue agent i receives only a local observation $o_i^t \in \Omega_i$ derived from telemetry within its assigned subnet. These observations include host-level security alerts, service status indicators, and firewall connectivity states but do not reveal the full global network state.

Agents select defensive actions $a_i^t \in A_i$ at every timestep, where the available actions include monitoring and analysis operations, remediation actions such as restoring compromised hosts, and connectivity control actions that modify firewall policies between network zones. The joint action $a_t = (a_1^t, \dots, a_n^t)$ induces a stochastic state transition according to

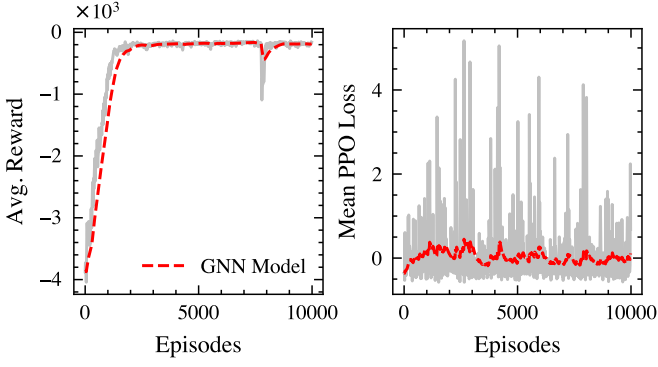


Fig. 2: Training results for MARL model. (Left) Average episode reward over training. (Right) Mean PPO training loss (linear scale). Solid lines indicate exponential moving averages ($\alpha = 0.05$); faint lines show raw values.

the environment dynamics and produces a scalar reward r_t that reflects mission impact, service availability, and adversarial activity. Operational priorities change across mission phases, which modify the reward penalties associated with failures in different network zones.

For evaluation, each blue policy is executed over 100 independent episodes with a horizon of $T = 500$ timesteps per episode. To assess robustness under heterogeneous adversarial behavior, we evaluate agents against six attacker strategies implemented in the CybORG environment: *Default*, *Stealth*, *Lateral*, *Impact*, *Deception*, and *Discovery*. Performance is measured using cumulative episodic reward $R = \sum_{t=1}^T r_t$, averaged across all episodes for each attacker strategy.

B. Agents & Robustness Analysis

We evaluate three types of blue agents: an expert-informed heuristic defense policy, a graph-based MARL agent, and a non-learning sleep agent, that functions as a baseline. The heuristic agent represents a rule-based defensive strategy derived from expert knowledge. It processes alerts generated from host and network telemetry and maintains queues of candidate remediation actions associated with observed security events. Alerts are categorized by their semantic type (connection, process, or file-related), and the agent applies a fixed priority hierarchy to determine the next action [10]. High-confidence indicators of compromise trigger remediation actions such as *Restore* or *Remove*, while lower-confidence alerts initiate *Analyse* operations to gather additional evidence. This structured decision process results in a reactive-proactive defense strategy that prioritizes restoring compromised hosts, removing malicious artifacts, and deploying decoys before executing monitoring operations.

The GNN agent follows the inductive graph-based MARL architecture introduced in [10]. Each blue agent observes its local network segment as an attributed graph $G_i = (V_i, E_i, X_i)$ representing hosts, services, and network relations within its responsibility zone. Node features encode security-relevant telemetry such as host properties, alerts, and firewall states.

A GNN computes node embeddings by iteratively aggregating information from neighboring nodes, allowing the policy to capture both local host attributes and relational dependencies in the network topology. Let $H^{(0)} = X_i$ denote the initial feature matrix and $\tilde{A} = A + I$ the adjacency matrix with self-loops. Node embeddings are computed as $H^{(\ell+1)} = \sigma\left(D^{-\frac{1}{2}}\tilde{A}D^{-\frac{1}{2}}H^{(\ell)}W^{(\ell)}\right)$, where $W^{(\ell)}$ are trainable weights, D is the degree matrix, and $\sigma(\cdot)$ is a nonlinear activation function. The resulting embeddings are shared between actor and critic networks trained using Proximal Policy Optimization (PPO). Actions are defined over graph entities rather than fixed indices, allowing the policy to select node-level actions such as *Analyse* or *Restore* on hosts and edge-level actions such as traffic control between subnetworks. This inductive formulation enables the policy to generalize across networks with varying topologies and numbers of entities without architectural modification. As a lower-bound baseline, we additionally evaluate a Sleep agent that performs no defensive actions.

To evaluate robustness under heterogeneous adversarial behavior, the blue agents are tested against multiple attacker strategies implemented as variants of the CybORG *FiniteStateRedAgent*. These attacker profiles modify the internal transition probabilities and host-selection heuristics of the adversary, resulting in distinct attack behaviors ranging from reconnaissance-oriented discovery strategies to aggressive privilege escalation and service disruption attempts.

Beyond mean reward metrics, we analyze agent behavior using reward and action decomposition. Reward decomposition [12] separates the total episodic reward into interpretable components corresponding to operational effects in the environment, including penalties caused by local work failures, access service failures, red impact actions, and action costs across subnetworks and mission phases (Table II). By aggregating these components across episodes, we obtain a detailed view of how different environmental factors contribute to agent performance.

Complementarily, action decomposition records the frequency with which each defensive action is selected during evaluation. Actions executed at each timestep are aggregated into empirical action distributions, enabling comparison of behavioral patterns across agents and attacker strategies. Together, reward and action decomposition provide interpretable diagnostics of how the evaluated policies respond to environmental incentives and how defensive strategies differ between heuristic and learning-based agents.

We define a robustness score that jointly captures expected performance, stochastic variability within individual attacker strategies, and sensitivity to adversarial strategy changes. Let $\mathcal{R}$ denote the set of considered red attacker strategies. For each blue agent b and attacker strategy $r \in \mathcal{R}$, we denote the mean episodic reward by $\mu_{b,r}$ and the corresponding standard deviation across evaluation episodes by $\sigma_{b,r}$.

We first compute the average performance and average

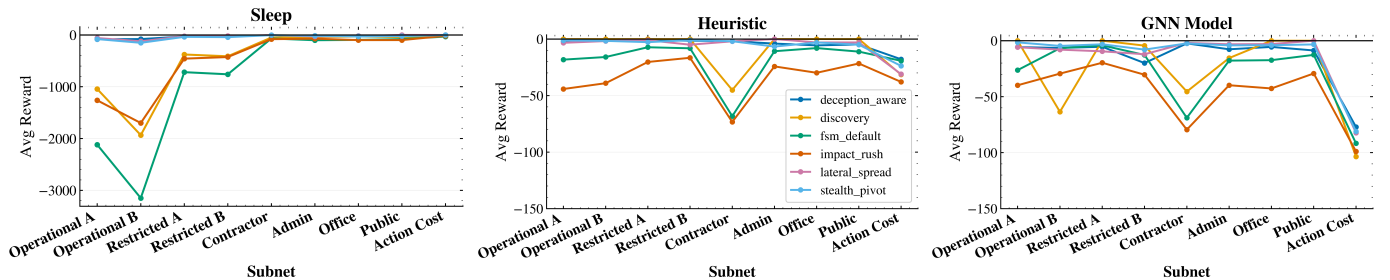


Fig. 3: Each subplot corresponds to one model and each line represents an attacker strategy. The x-axis shows subnet-level reward contributions and action cost, while the y-axis shows the corresponding reward.

such as `Restore` or `Remove`. This behavior reflects the expert-designed strategy underlying the heuristic policy, which prioritizes situational awareness and evidence gathering before performing disruptive interventions.

The GNN agent, in contrast, relies heavily on connectivity-control actions such as `AllowTrafficZone` and `BlockTrafficZone`. While these actions influence network connectivity, they do not necessarily remove compromised hosts or eliminate attacker persistence mechanisms. In some cases, excessive traffic blocking can even degrade network functionality, generating penalties due to disrupted legitimate activity. At the same time, analysis-related actions appear significantly less frequently in the learned policy. This indicates that the agent does not sufficiently exploit information-gathering actions to reduce uncertainty before executing defensive interventions. As a result, the policy may operate with incomplete situational awareness, leading to inefficient or unnecessary remediation actions.

D. Robustness Analysis

To evaluate agent robustness under adversarial strategy variation, we compute the robustness score defined in Section III. This metric combines three components: average performance across attacker strategies, within-strategy reward variability, and cross-strategy sensitivity. Table III shows that the heuristic agent achieves the highest robustness score. This result indicates not only strong average performance but also relatively stable behavior across both stochastic episodes and different attacker strategies.

The GNN agent achieves the second-best robustness score but exhibits significantly higher variability. Both the within-strategy standard deviation and the cross-strategy sensitivity

are larger compared to the heuristic agent. This confirms that the learned policy is more sensitive to variations in attacker behavior. The Sleep baseline performs substantially worse than both active defense agents. This result confirms that active defensive interventions are essential for maintaining acceptable network resilience in the CC4 environment.

V. CONCLUSION & FUTURE WORK

This paper evaluated the robustness of heuristic and graph-based MARL cyber defense agents under varying attacker strategies in the Cyborg CAGE Challenge 4 environment. By combining performance evaluation with reward and action decomposition, we analyzed both overall performance and the underlying decision-making behavior of the agents. The results show that the expert-informed heuristic agent achieves the highest robustness, maintaining stable performance across heterogeneous attacker behaviors, while the learned GNN policy exhibits higher variability and stronger sensitivity to adversarial strategy changes.

We consider a single representative GNN-based MARL agent to enable a controlled comparison under identical conditions, focusing on identifying characteristic failure modes of learned policies under adversarial variation rather than broad generalization. Behavioral analysis reveals that the learned policy over-relies on connectivity-control actions and underutilizes analysis actions, leading to less effective containment of attacker activity in mission-critical zones. These weaknesses can primarily be attributed to (i) reward misalignment, (ii) insufficient use of information-gathering actions, and (iii) over-reliance on connectivity control.

Overall, the results highlight that robustness in ACD cannot be assessed solely through aggregate reward metrics, but requires systematic evaluation across diverse attacker behaviors combined with interpretable behavioral diagnostics.

A. Future Work

Future work should focus on improving reward design, training procedures, and evaluation breadth for learning-based ACD agents. In particular, aligning rewards more closely with adversarial activity can reduce noise and mitigate reward misalignment. Encouraging information-gathering behavior

TABLE III: Robustness score incorporating both within-strategy variability ($\bar{\sigma}$) and cross-strategy sensitivity ($\text{Std}_r(\mu_{b,r})$). $R(b) = \bar{\mu} - \bar{\sigma} - \text{Std}_r(\mu_{b,r})$. Higher (less negative) is better.

Model	$\bar{\mu}$	$\bar{\sigma}$	$\text{Std}_r(\mu_{b,r})$	$R(b)$
Heuristic	-96.29	33.81	91.75	-221.84
SimpleGNN	-168.10	51.73	104.92	-324.75
Sleep	-2612.82	578.98	2636.01	-5827.81

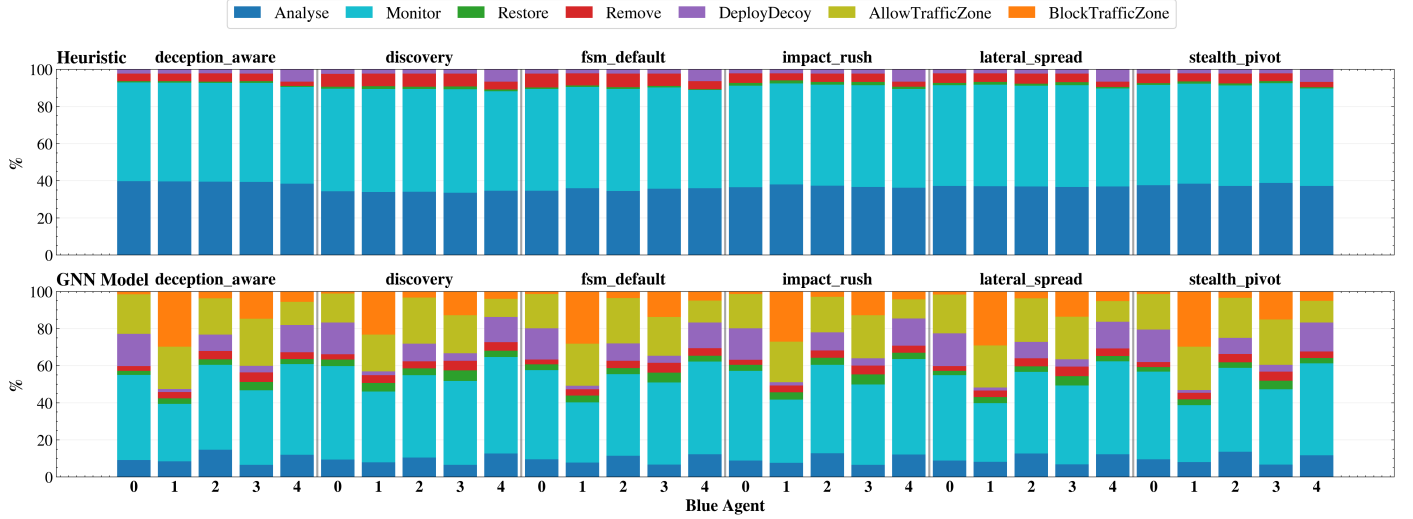


Fig. 4: Action decomposition of the blue agents across attacker profiles. Each group corresponds to one attacker strategy and contains five bars representing the blue agents (0–4). Bars are stacked by action type and show the percentage of actions executed during evaluation. Results are shown for the heuristic and MARL model.

(e.g., via intrinsic rewards) and regularizing connectivity-control actions may lead to more balanced and effective defensive strategies.

Robustness-aware training, such as curriculum learning or adversarial training across diverse attacker strategies, is a promising direction to improve generalization and reduce sensitivity to distributional shifts. Incorporating expert knowledge through heuristic-guided DRL or imitation learning may further enhance stability and sample efficiency.

Finally, future work should extend the evaluation to a broader range of MARL approaches, architectures, and training setups to disentangle algorithmic from design-specific limitations. In addition, robustness can be incorporated directly into training objectives, for example by optimizing worst-case performance or penalizing large performance variance, supported by explainability methods such as SHAP.

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